

MATH 3310 HOMEWORK ASSIGNMENT 12

DUE ON FRIDAY 2 DECEMBER 2022

Please turn in your homework in class on Thursday or to Weeks Hall 347 no later than 11:30 am on Friday.

- (1) For each of the next functions $f: \mathbb{R} \rightarrow \mathbb{R}$ decide if it is 1-to-1 and/or onto. You must justify your answers.

(a) $f(x) = x^3 - 2x + 7$.

(b) $f(x) = e^x + x$.

(c) $f(x) = 3x - e^x$.

- (2) Consider the function $f: \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{1}{x-1}.$$

- (a) Prove that f is injective.
(b) Determine the range of f .
(c) The corestriction $f: \mathbb{R} - \{1\} \rightarrow \text{range}(f)$ has an inverse, find it.
- (3) Let A , B , and C be sets and consider functions $f: A \rightarrow B$ and $g: B \rightarrow C$. Prove that if the composite $g \circ f: A \rightarrow C$ is onto, then g is onto.
- (4) Decide which of the statements (a)–(d) below that are proved by the following argument. Justify your answers.

Let A be a proper subset of B , that is $A \subseteq B$ and $A \neq B$, and assume that there exists a bijective function

$$f: A \rightarrow B.$$

Suppose B is finite, then A is finite as well, and since A is a proper subset of B one has $|A| < |B|$. The function f is 1-to-1, so one has $|f(A)| = |A|$. On the other hand, f is also onto, so one has $|f(A)| = |B|$. Contradiction!

- (a) A set B cannot have the same cardinality as a proper subset A of B .
(b) If a set B has the same cardinality as a proper subset A of B , then B is not finite.
(c) The cardinality of a proper subset A of a set B is less than the cardinality of B .
(d) If A is a proper subset of B , then there cannot exist a bijective function $f: A \rightarrow B$.
- (5) Let R be the relation on $\mathbb{Z}$ given by

$$a R b \text{ if } a^3 \equiv_4 b^3.$$

- (a) Prove that R is an equivalence relation.
(b) Determine all the equivalence classes for R .